 

# RESKO-SMOLSKO, Poland

WMO: **122100**

Lat: **53.7636N** Lon: **15.3933E** Elev: **53** StdP: **100.69** Time Zone: **1.00 (EUC)** Period: **00-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **-12.5** | **-9.4** | **-15.6** | **1.0** | **-10.9** | **-12.7** | **1.3** | **-8.0** | **9.6** | **5.3** | **8.6** | **5.1** | **1.4** | **70** | **0.515** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **10.2** | **29.2** | **19.5** | **27.3** | **18.8** | **25.6** | **18.1** | **20.8** | **26.8** | **19.9** | **25.4** | **19.0** | **23.7** | **3.0** | **150** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **18.7** | **13.6** | **22.9** | **18.0** | **13.0** | **22.0** | **17.2** | **12.3** | **21.2** | **60.3** | **27.0** | **57.2** | **25.3** | **54.2** | **23.5** | **24.9** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **7.6** | **6.5** | **5.7** | DB | **-16.2** | **32.9** | **5.0** | **2.1** | **-19.9** | **34.5** | **-22.8** | **35.7** | **-25.6** | **36.9** | **-29.3** | **38.5** | (4) |
| (5) | | | | WB | **-16.4** | **22.7** | **4.8** | **1.3** | **-19.9** | **23.7** | **-22.7** | **24.4** | **-25.4** | **25.2** | **-28.9** | **26.2** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | DBAvg | | **9.0** | **-0.6** | **0.5** | **3.5** | **8.6** | **12.7** | **15.9** | **18.1** | **18.1** | **13.9** | **9.4** | **5.2** | **1.8** | (6) |
| (7) | | DBStd | | **7.54** | **4.92** | **4.35** | **4.31** | **3.89** | **3.89** | **3.29** | **2.97** | **2.92** | **3.00** | **3.52** | **3.27** | **4.36** | (7) |
| (8) | | HDD10.0 | | **1342** | **328** | **266** | **203** | **71** | **16** | **1** | **0** | **0** | **3** | **54** | **147** | **253** | (8) |
| (9) | | HDD18.3 | | **3511** | **587** | **499** | **460** | **291** | **182** | **88** | **42** | **41** | **137** | **277** | **395** | **512** | (9) |
| (10) | | CDD10.0 | | **967** | **0** | **0** | **2** | **30** | **98** | **178** | **251** | **250** | **120** | **36** | **2** | **0** | (10) |
| (11) | | CDD18.3 | | **93** | **0** | **0** | **0** | **0** | **6** | **16** | **34** | **33** | **4** | **0** | **0** | **0** | (11) |
| (12) | | CDH23.3 | | **951** | **0** | **0** | **0** | **6** | **71** | **187** | **363** | **287** | **37** | **0** | **0** | **0** | (12) |
| (13) | | CDH26.7 | | **234** | **0** | **0** | **0** | **1** | **11** | **49** | **96** | **73** | **4** | **0** | **0** | **0** | (13) |
| (14) | Wind | WSAvg | | **2.6** | **2.9** | **2.9** | **2.9** | **2.6** | **2.4** | **2.4** | **2.3** | **2.3** | **2.3** | **2.5** | **2.7** | **3.1** | (14) |
| (15) | Precipitation | PrecAvg | | **702** | **54** | **44** | **51** | **36** | **54** | **71** | **78** | **82** | **64** | **52** | **56** | **61** | (15) |
| (16) | | PrecMax | | **938** | **126** | **85** | **97** | **103** | **89** | **177** | **170** | **171** | **147** | **124** | **122** | **109** | (16) |
| (17) | | PrecMin | | **578** | **1** | **8** | **11** | **11** | **10** | **28** | **2** | **18** | **17** | **22** | **3** | **14** | (17) |
| (18) | | PrecStd | | **96** | **31** | **20** | **24** | **20** | **21** | **29** | **43** | **36** | **29** | **27** | **25** | **27** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **10.6** | **11.5** | **17.3** | **23.7** | **28.2** | **31.1** | **31.9** | **31.7** | **27.4** | **20.8** | **14.1** | **11.8** | (19) |
| (20) | | | MCWB | | **8.7** | **8.6** | **10.5** | **14.5** | **18.3** | **20.5** | **20.8** | **20.7** | **18.8** | **15.3** | **12.4** | **10.3** | (20) |
| (21) | | 2% | DB | | **8.4** | **9.4** | **14.0** | **20.6** | **25.4** | **27.8** | **29.2** | **28.5** | **24.1** | **18.4** | **12.1** | **9.9** | (21) |
| (22) | | | MCWB | | **7.0** | **7.2** | **8.7** | **12.6** | **16.7** | **18.9** | **19.7** | **19.3** | **16.7** | **14.4** | **10.3** | **8.7** | (22) |
| (23) | | 5% | DB | | **7.1** | **7.8** | **11.6** | **18.4** | **22.8** | **25.3** | **27.2** | **26.4** | **21.9** | **16.5** | **10.9** | **8.6** | (23) |
| (24) | | | MCWB | | **6.0** | **6.0** | **8.2** | **11.4** | **15.3** | **18.2** | **19.0** | **18.4** | **15.9** | **13.5** | **9.4** | **7.4** | (24) |
| (25) | | 10% | DB | | **5.7** | **6.3** | **9.7** | **16.2** | **20.6** | **23.1** | **25.2** | **24.4** | **19.7** | **14.8** | **9.8** | **7.2** | (25) |
| (26) | | | MCWB | | **4.7** | **4.7** | **7.0** | **10.4** | **14.3** | **17.1** | **18.3** | **17.9** | **15.0** | **12.5** | **8.6** | **6.0** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **9.2** | **9.4** | **11.4** | **15.7** | **19.6** | **22.3** | **21.9** | **21.9** | **19.8** | **16.3** | **13.0** | **10.6** | (27) |
| (28) | | | MCDB | | **10.6** | **10.8** | **15.1** | **22.4** | **25.8** | **28.8** | **28.8** | **28.7** | **25.0** | **19.4** | **14.0** | **11.5** | (28) |
| (29) | | 2% | WB | | **7.2** | **7.6** | **9.8** | **13.3** | **17.7** | **20.0** | **20.8** | **20.6** | **17.6** | **14.9** | **10.7** | **8.9** | (29) |
| (30) | | | MCDB | | **8.2** | **9.0** | **12.7** | **19.2** | **23.2** | **26.4** | **27.1** | **26.2** | **22.2** | **17.5** | **11.7** | **9.8** | (30) |
| (31) | | 5% | WB | | **6.0** | **6.1** | **8.3** | **12.0** | **16.4** | **18.9** | **19.9** | **19.6** | **16.5** | **13.8** | **9.6** | **7.5** | (31) |
| (32) | | | MCDB | | **6.9** | **7.5** | **11.0** | **17.2** | **21.4** | **24.0** | **25.6** | **24.3** | **20.3** | **16.0** | **10.6** | **8.5** | (32) |
| (33) | | 10% | WB | | **4.8** | **4.7** | **7.2** | **10.8** | **14.9** | **17.6** | **19.0** | **18.7** | **15.5** | **12.7** | **8.7** | **6.2** | (33) |
| (34) | | | MCDB | | **5.8** | **6.2** | **9.5** | **15.4** | **19.3** | **22.0** | **23.7** | **22.6** | **19.1** | **14.6** | **9.7** | **7.0** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **4.3** | **5.3** | **7.5** | **10.1** | **10.9** | **10.6** | **10.2** | **10.2** | **9.1** | **7.1** | **4.5** | **3.9** | (35) |
| (36) | | 5% DB | MCDBR | | **4.2** | **6.5** | **10.8** | **14.1** | **14.3** | **14.4** | **14.4** | **14.0** | **12.8** | **9.7** | **5.5** | **4.7** | (36) |
| (37) | | | MCWBR | | **3.8** | **4.9** | **6.6** | **7.6** | **7.2** | **7.5** | **6.2** | **6.3** | **6.4** | **6.2** | **4.3** | **4.3** | (37) |
| (38) | | 5% WB | MCDBR | | **4.2** | **5.8** | **9.3** | **12.7** | **12.7** | **12.9** | **12.5** | **11.5** | **11.0** | **8.9** | **5.1** | **4.7** | (38) |
| (39) | | | MCWBR | | **3.9** | **4.8** | **6.2** | **7.2** | **6.9** | **7.2** | **6.0** | **5.9** | **6.2** | **6.1** | **4.3** | **4.5** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.312** | **0.337** | **0.369** | **0.407** | **0.405** | **0.401** | **0.424** | **0.421** | **0.381** | **0.361** | **0.339** | **0.309** | (40) |
| (41) | | taud | | | **2.301** | **2.299** | **2.265** | **2.224** | **2.279** | **2.317** | **2.268** | **2.286** | **2.376** | **2.391** | **2.357** | **2.307** | (41) |
| (42) | | Ebn at Noon | | | **643** | **749** | **803** | **819** | **842** | **849** | **819** | **797** | **786** | **715** | **605** | **568** | (42) |
| (43) | | Edh at Noon | | | **60** | **82** | **105** | **124** | **124** | **121** | **125** | **116** | **94** | **75** | **57** | **49** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **0.54** | **1.20** | **2.42** | **4.17** | **5.25** | **5.55** | **5.13** | **4.32** | **3.14** | **1.62** | **0.66** | **0.38** | (44) |
| (45) | | RadStd | | | **0.06** | **0.17** | **0.27** | **0.55** | **0.61** | **0.43** | **0.65** | **0.37** | **0.34** | **0.23** | **0.09** | **0.04** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (36 neighbors) | **+0.40** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **-140** | **+60** | **N/A** | (47) |

Nomenclature: See separate page